Using Machine Learning (XGBoost) Methods to Analyze Traffic Flows in Lima, Peru

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February 5, 2026

As the global population grows exponentially, an increasing proportion of the world has begun to live in urban environments, with an estimated [%] living in cities by [] [?]. Accurate traffic flow prediction models are necessary in order to optimize traffic patterns and inform sustainable, safe, and cost-effective infrastructure. Recent advances in machine learning (ML) and deep learning techniques enhance Intelligent Transportation Systems (ITS) by more accurately predicting traffic flows than historical averages or linear regression techniques. However, the majority of studies concentrate on highly-developed cities where data is readily available, while fewer studies focus on the implementation of ML-powered traffic control monitoring systems in developing cities. Since developing cities are likely to suffer from uneven development which begets traffic congestion and traffic-related accidents, understanding traffic flows in developing cities is crucial for policymakers and urban planners. This project aims to create a dynamic traffic flow model of Lima, Perú, utilizing Extreme Gradient Boosting (XGBoost) and Random Forest (RF) machine learning techniques. XGBoost and RF were chosen due to their relatively low computational requirement compared to deep learning neural networks, their high performance with limited data, and their proficiency with tabular data. A variety of studies have displayed the effectiveness of XGBoost and RF in modeling traffic flows in developing countries compared to ANN, LSTM, or SVM models [?] [?]. Open-Source Map (OSM) data will be used to generate a road map of the city, while additional open-source data from the Autoridad de Transporte Urbano (ATU), Indicadores Mensuales de Tráfico, and TomTom will be used to model population flows according to spaciotemporal indicators.

In your proposal, you will provide justification on why your project matters based on work which has been done in this area, using in-line citations [?] to refer to existing works. Include what your project will accomplish and how your software will function.

When you use an image, such as in Figure 1, refer to it in the text.

Figure 1: Archie

Appendix

Features of the project

- Visualization of the statistical baseline (found using historical average linear regression techniques)
- Table of XGBoost results
- Table of RF results
- Plot to display the importance of each feature on the outcome (calculated using SHAP, PFI, and/or FIRM)
- Loss plot to determine the accuracy of each model
- Map results to GIS to create a color-coded visualization of congestion on each street
- Implement capacity of GIS model to vary depending on time, date, and weather features.
- Implement capacity of GIS model to respond to conditions such as a closed road or or an additional freeway.

References